TDA for Audio Deepfake Detection

A Technical Proposal for Explainable Forensic Classification

Prepared for: CLTC / TDA Course Project Discussion

Executive Summary

We propose a class project that uses persistent homology to detect audio deepfakes in a way that is not only accurate but explainable. The approach constructs a point cloud from an audio sample by computing physically motivated features—MFCCs and their time derivatives—in sliding windows, then applies Vietoris-Rips persistent homology to detect structural anomalies in the resulting trajectory. What makes this more than a standard TDA classification exercise is a dimensional ablation mechanism: when a sample is flagged, we systematically remove groups of features and recompute PH to isolate which physical property of speech (e.g., spectral continuity, pitch dynamics) drives the anomaly. This converts a binary detection into a traceable explanation.

The minimal class deliverable is focused: train and evaluate on the ASVspoof 2019 Logical Access dataset, using a 39-dimensional MFCC embedding (static + Δ + Δ²), computing H_0 and H_1 via Ripser, vectorizing with persistence images, and classifying with an SVM. One targeted ablation experiment—removing the continuity-related Δ/Δ² dimensions and measuring the effect on both classification and topological signatures—tests the core hypothesis. Success does not require beating state-of-the-art neural detectors; it requires demonstrating that topological features derived from physically motivated embeddings carry discriminative and interpretable signal.

The central research risk is empirical: it is possible that the topological signatures of real and synthetic speech do not differ sufficiently in MFCC space to support reliable discrimination. This is what the project tests. The broader research direction—modular extensibility, additional feature dimensions, adversarial robustness, potential generalization to other modalities—is described in this document to provide context, but the class project scope is deliberately narrow.

1. Problem Statement

Audio deepfakes generated by modern synthesis systems (VALL-E, RVC, So-VITS-SVC, XTTS, and others) are increasingly difficult to distinguish from genuine speech. Current detection methods rely primarily on neural network classifiers that output a confidence score but provide no explanation of why a sample was flagged. Even architectures that incorporate attention mechanisms or spectro-temporal graph networks such as AASIST [1] are fundamentally score-producing systems: they can indicate that something is anomalous, but they cannot articulate which physical property of the speech signal is violated.

This proposal describes a detection methodology built on Topological Data Analysis (TDA), specifically persistent homology, that aims to be both discriminative and explainable. The core idea is not simply to apply TDA as another classification layer, but to use it as the backbone of a framework where each detection can be traced to a specific, physically meaningful property of the audio signal. Whether this approach yields competitive detection accuracy, or whether the topological signatures of real and synthetic speech are sufficiently different to support reliable discrimination, are empirical questions this project would investigate.

2. Core Insight: Physically Motivated Embeddings

Raw audio is essentially a one-dimensional signal. Persistent homology is most powerful when applied to data that naturally inhabits a moderate-to-high-dimensional space with meaningful geometric structure. Naively inflating the dimensionality of an audio signal—for example, through arbitrary time-delay embeddings—produces point clouds where any topological features detected may be artifacts of the embedding choices rather than properties of the underlying signal.

Our approach attempts to resolve this by constructing embeddings where every dimension corresponds to a specific, physically grounded property of human speech production. The vocal tract is a physical system with known constraints: articulatory inversion studies using electromagnetic articulography (EMA) show that measured articulator trajectories are strongly low-pass, with 99% of energy contained below approximately 21 Hz across all tracked articulators [2]. This means real speech articulation—tongue, lips, jaw, velum—cannot change arbitrarily fast, and therefore acoustic correlates driven by articulation (formant movement, spectral envelope evolution) inherit bounded temporal variation—though not always in a trivially smooth way due to nonlinear source-filter interactions and coarticulation. Similarly, studies of voluntary pitch movements have quantified physiological limits on pitch-change speed, reporting maximum rates on the order of 67–96 semitones per second depending on direction and excursion size [3].

The central hypothesis is this: if we construct a point cloud in a feature space where each axis represents a property that real speech must satisfy due to physical constraints, then real and synthetic speech may trace trajectories with different topological signatures in that space. Persistent homology gives us the tools to detect and quantify those differences—if they exist. Validating this hypothesis is the primary research objective.

3. Staging the Argument: Design Principle, Empirical Hypothesis, Interpretability Objective

It is important to distinguish three claims that this proposal makes, because they differ in their current evidentiary status:

Claim 1 (Design Principle): Each embedding dimension should correspond to a physically meaningful property of speech production, grounded in known biomechanical or acoustic constraints. This is a methodological commitment, not an empirical claim. It is justified by the observation that arbitrary dimensionality inflation degrades persistent homology on finite, noisy point clouds (see Section 4), and by the principle that forensic tools should be built on testable, interpretable foundations.

Claim 2 (Empirical Hypothesis): Persistent homology computed on physically motivated embeddings will yield topological signatures that differ between real and synthetic speech in a way that supports reliable classification. This is the core research question. It is plausible—deepfake systems may fail to perfectly replicate the constrained dynamics of real speech—but it is not established. The project would test this.

Claim 3 (Interpretability Objective): When a topological anomaly is detected, dimensional ablation can isolate which feature groups drive the anomaly, producing an explanation in terms of specific violated physical properties. This is the most ambitious claim. Even if Claims 1 and 2 hold, it is possible that the topological signal is distributed across many dimensions in a way that resists clean attribution. Ablation studies would probe whether localized explanations are achievable.

Staging the argument this way clarifies what the project assumes, what it tests, and what it aspires to.

4. Why Dimensionality Matters and Why We Must Be Deliberate

A natural question is whether we can simply add as many feature dimensions as possible and let the topology sort itself out. In theory, the homology of an underlying manifold is invariant under embedding into higher-dimensional ambient spaces. In practice, persistent homology is computed not on the true manifold but on finite, noisy point clouds via distance-based constructions (typically Vietoris-Rips complexes). Recent theoretical work confirms that this distinction has serious consequences.

4.1 Distance Distortion and Concentration

Every added dimension contributes to pairwise Euclidean distances. If an added dimension contains noise or variation unrelated to the real/fake distinction, it inflates and distorts distances between points. Fernandez et al. [4] provide a direct characterization of this failure mode: as ambient dimension grows, distances become dominated by noise coordinates, birth/death behavior in persistence diagrams shifts, and all distances become too similar, hiding true topological features. The authors demonstrate that the Vietoris-Rips filtration can fail to recover known topology even at moderate noise levels when ambient dimensionality is high.

4.2 Spurious Features

Irrelevant dimensions that contain any structured variation (as real-world features almost always do) can create apparent topological features—holes, connected components, higher-dimensional voids—that are artifacts of noise rather than reflections of the underlying data-generating process.

4.3 Mitigation Strategies

Fernandez et al. [4] suggest that kNN-graph-based spectral distances (effective resistance, diffusion distance) can recover topology more robustly in high-dimensional noisy settings than Euclidean distances. Additionally, Sheehy [5] shows that Johnson–Lindenstrauss-style random projection into O(log n) dimensions can preserve persistent homology of distance functions up to a bounded distortion factor, providing a principled dimensionality reduction route with theoretical guarantees—a relatively rare property compared to more heuristic methods like UMAP.

4.4 Practical Implication

Each dimension in our embedding must earn its place. We include a feature dimension only when we have a principled physical or signal-theoretic argument that it captures a property relevant to distinguishing real from synthetic speech. This constraint is not a limitation—it is both the source of the method's potential explanatory power and a methodological necessity given the sensitivity of PH to ambient noise.

5. Proposed Embedding Strategy

We construct a point cloud from the audio signal using a sliding-window approach. For each short, overlapping time window, we compute a feature vector. The trajectory of these feature vectors through the resulting space forms the point cloud on which we compute persistent homology.

5.1 Base Features: MFCCs and Derivatives

Mel-frequency cepstral coefficients (MFCCs) are a standard audio representation that compresses the spectral envelope of a signal into a compact vector (typically 13 coefficients). MFCCs approximate the shape of the vocal tract filter as estimated from the signal's short-time spectrum [6, 7]. Changes in MFCC values over time reflect changes in vocal tract configuration.

We include three levels of MFCC-derived features per window:

Feature	Dimensions	Physical Interpretation
MFCCs (static)	13	Vocal tract configuration at each instant
ΔMFCCs (first derivative)	13	Rate of vocal tract change—captures transition dynamics
Δ^2MFCCs (second derivative)	13	Acceleration of vocal tract change—captures smoothness and continuity

This gives a 39-dimensional base embedding per window. The delta and delta-delta coefficients are particularly important: they explicitly encode the continuity and smoothness of spectral evolution. Real speech, produced by a physical articulatory system whose trajectories are strongly low-pass (99% of energy below ~21 Hz [2]), should exhibit characteristic smoothness in these derivative features. The strongest defensible claim is not that MFCCs uniquely parameterize vocal tract geometry, but that they are a compact representation of spectral envelope shape, and when augmented with deltas they encode how that envelope changes over time—which is influenced by articulatory motion constraints and coarticulation [7].

An important caveat: recent forensic work by Kelly et al. [8] has argued that MFCCs alone can have weak evidential strength for deepfake detection in certain conditions, while segmental, articulator-linked features (e.g., vowel formants) show stronger discriminative value. This supports the design decision to use MFCCs as one physically motivated axis among several, and to use ablation to empirically demonstrate which axes carry the actual discriminative signal.

5.2 Additional Physically Motivated Dimensions

Beyond MFCCs, additional features can be selectively included, each with a specific physical justification and, where available, quantified normative bounds:

Feature	Dims	Physical Basis	Known Constraints / Notes
F0 (fundamental frequency)	1–2	Vocal fold vibration rate	Bounded pitch-change speed: ~67–96 st/s max [3]
F0 slope / acceleration	1–2	Laryngeal dynamics	Exceeding physiological limits motivates synthetic origin hypothesis
Spectral flux	1	Frame-to-frame spectral change	Smoothness bounded by articulatory inertia [2]
Jitter	1	Cycle-to-cycle period variability	Typical range ~0.3–0.7% for sustained vowels [9]; oversmoothing is a synthesis tell
Shimmer	1	Cycle-to-cycle amplitude variability	Typical range ~1–3% for sustained vowels [9]; absence suggests synthetic regularity
Harmonic-to-noise ratio	1	Periodic vs. aperiodic energy	Voice quality and breathiness; synthesis may produce unnaturally clean signals
Formant frequencies (F1–F3)	3–6	Vocal tract resonances	F1 ≈ tongue height; F2 ≈ tongue frontness [8]; deepfakes may fail individual realizations
Formant bandwidths	3	Damping of tract resonances	Physical consistency of vocal tract modeling

The total embedding dimensionality might range from 39 (MFCCs only) to approximately 55–60 with all additional features included. This is within the range where persistent homology on point clouds of a few hundred to a few thousand points remains computationally tractable. The formant and jitter/shimmer features connect more directly to articulatory physiology than MFCCs do, which would strengthen the explanatory narrative if ablation implicates those dimensions.

5.3 Modularity: Adding Dimensions Over Time

A key advantage of this framework is that the embedding is modular and extensible. If a new generation of deepfake systems learns to match existing feature constraints, we can investigate what new physical property they are violating, design a feature that captures it, add it as a new dimension, and validate empirically that it improves topological discrimination. The framework grows without needing to be rebuilt.

6. The TDA Pipeline

Given the embedded point cloud, the topological analysis proceeds as follows.

6.1 Point Cloud Construction

From an audio clip of N frames, with a sliding window of width w and step size s, we obtain approximately (N − w) / s points in our d-dimensional feature space. For a typical 10-second clip at standard audio processing parameters, this yields on the order of several hundred to a few thousand points.

6.2 Persistent Homology Computation

We compute persistent homology on this point cloud using the Vietoris-Rips filtration. At each filtration parameter ε, we form a simplicial complex by connecting points within distance ε and track the birth and death of topological features across dimensions:

- **H_0 (connected components):** Captures the clustering structure of the trajectory. If real speech with smooth transitions produces a trajectory that connects into a single component at relatively small ε, synthetic speech with discontinuities might show fragmented clusters that persist longer before merging. This is a hypothesis to be tested.
- **H_1 (loops / 1-cycles):** May capture cyclical or repetitive structure. Real speech has natural periodicities (prosodic contours, syllable production) that could produce characteristic loops in feature space. Whether synthetic speech produces loops with measurably different persistence is an empirical question.
- **Higher homology (H_2+):** May capture higher-order structural properties, though signal at these levels is typically weaker for moderate point cloud sizes. Initial work should focus on H_0 and H_1.

6.3 Precedent: TDA on Audio Representations

There is credible precedent for applying persistent homology to audio data. Tulchinskii et al. [10] apply TDA to speech processing and model introspection at INTERSPEECH 2023, deriving topological features from transformer attention maps and embeddings and demonstrating performance gains on classification tasks. More directly relevant, Cuevas et al. [11] at Spotify Research describe “Topological Fingerprints for Audio Identification,” applying persistent homology via filtered cubical complexes computed from mel spectrograms, encoding audio content via local Betti curves, and reporting robustness to obfuscations including time stretching and pitch shifting. While this is an identification task rather than deepfake detection, it demonstrates that PH on audio-derived spectral representations can carry stable, meaningful information under transformations that often break naive methods.

An alternative to Vietoris-Rips on point clouds is cubical-complex PH computed directly on 2D spectrogram arrays, as in the Spotify work. This may be computationally favorable and could align naturally with recent deepfake detection research identifying artifacts in spectrogram magnitude texture [12]. This is worth considering as an alternative or complementary filtration approach.

6.4 Topological Summary Representations

The raw output of persistent homology is a persistence diagram: a multiset of (birth, death) pairs for each homological dimension. For downstream classification, we convert these into vectorized representations:

- **Persistence images:** Discretize the persistence diagram into a weighted, smoothed grid. Produces a fixed-size vector suitable for standard classifiers.
- **Persistence landscapes:** A functional summary that converts the diagram into a sequence of piecewise-linear functions, which can be averaged and compared statistically.
- **Summary statistics:** Total persistence, maximum persistence, entropy of the diagram, Betti curves at selected ε values.

These representations become the feature input to a classifier (e.g., SVM, random forest, or logistic regression—deliberately simple to preserve interpretability).

7. The Explainability Mechanism: Dimensional Ablation

This is the most distinctive aspect of the proposed methodology and what separates it from applying TDA as just another feature extraction step.

7.1 The Procedure

When a sample is flagged as a potential deepfake, we do not simply report a classification label. We run an ablation analysis:

1. Compute the full topological signature using all embedding dimensions.
2. Systematically remove groups of related dimensions (e.g., remove all continuity-related features, or all formant features, or jitter/shimmer).
3. Recompute persistent homology after each removal.
4. Compare: which removal causes the topological distinction between this sample and the reference distribution of real speech to collapse or significantly change?

7.2 What This Could Produce

If the approach works as hypothesized, the result would be a specific, traceable explanation. For example:

> *"The topological signature of this audio sample deviates significantly from the reference distribution of authentic speech. Ablation analysis suggests that the deviation is primarily concentrated in the dimensions corresponding to spectral continuity (ΔMFCC and Δ²MFCC features). The persistence of H_0 components in these dimensions is elevated, suggesting fragmentation in spectral transition dynamics that may be inconsistent with physical vocal tract constraints. Removing the continuity-related dimensions substantially reduces the topological anomaly, indicating that the detection is primarily driven by discontinuity in spectral evolution rather than by static spectral properties."*

Important caveat: ablation can demonstrate statistical dependence of an anomaly on a feature group. It does not by itself confirm that a particular physical law was violated—it shows that the anomaly is localized to dimensions associated with that property. The strength of the causal interpretation depends on how well-grounded the feature's connection to physiology is. This is why the physical motivation of each embedding dimension matters: it is what converts a statistical finding ("the anomaly is in these dimensions") into a substantive claim ("spectral continuity appears to be violated").

7.3 Computational Cost

The ablation analysis is more expensive than a single classification pass because it requires multiple recomputations of persistent homology. However, this cost is incurred only when a sample has already been flagged. The expensive explanatory analysis triggers only on positives—which, in a well-calibrated system, should be a small fraction of total samples.

8. Building Empirical Backing Through Ablation Studies

The ablation mechanism also serves a scientific purpose beyond individual-sample explanation. By systematically studying which feature dimensions drive topological discrimination across a corpus, we can build empirical evidence for or against specific hypotheses:

- If removing continuity features (Δ and Δ^2 MFCCs) consistently collapses the distinction, that constitutes evidence that current deepfakes systematically violate spectral continuity constraints.
- If removing jitter/shimmer features has a similar effect, that suggests deepfakes fail to replicate natural vocal fold micro-perturbations—consistent with the known tendency of synthesis pipelines to produce excessively regular excitation [9].
- If removing formant-related features collapses discrimination, that aligns with forensic research showing deepfakes can fail to capture speaker-specific segmental realizations [8].
- If no single feature group drives discrimination, that is also informative: it suggests the topological anomaly reflects a global rather than localized constraint violation.
- If some feature groups contribute nothing, we can prune them, reducing dimensionality and improving the quality of the topological computation per Section 4.

This transforms the system from a pure classifier into a hypothesis-testing framework.

9. Extensibility and Adversarial Adaptation

Deepfake generation is an adversarial arms race. Published work demonstrates that attackers can explicitly optimize against detector features—for example, the Malafide attack [13] uses adversarial convolutive noise to degrade countermeasure performance by an order of magnitude via channel-like filtering. As generators improve, some current detection signals will degrade. The modular embedding design provides a natural response:

1. Observe that a new generation of deepfakes passes existing tests.
2. Investigate via ablation: which feature dimensions no longer discriminate?
3. Hypothesize: what new physical constraint are these systems still violating?
4. Design a new feature dimension that captures that constraint.
5. Add it to the embedding and validate empirically.

This cycle mirrors how forensic science generally evolves. The approach also establishes a methodology template that could generalize beyond audio to video (biomechanical constraints) or images (sensor physics), though those extensions are substantially more complex and beyond the scope of this project.

10. Implementation Overview

10.1 Software and Libraries

Component	Tool / Library	Notes
Audio feature extraction	librosa (Python)	MFCCs, delta/delta-delta, F0, spectral features
Additional voice features	parselmouth / Praat	Jitter, shimmer, HNR, formants
Persistent homology	Ripser [14] or GUDHI [15]	Efficient Vietoris-Rips; GUDHI also supports cubical complexes
Topological vectorization	Giotto-tda [16] or persim	Persistence images, landscapes; Giotto-tda is scikit-learn-compatible
Classification	scikit-learn	SVM, random forest, logistic regression
Evaluation	scikit-learn + custom	Cross-validation, calibration, error rate analysis

10.2 Computational Requirements

For point clouds of 500–3000 points in 40–60 dimensions, Vietoris-Rips persistent homology up to H_1 is tractable on standard hardware using Ripser [14]. A single audio clip can typically be processed in seconds to low minutes. Batch processing is parallelizable. If dimensionality becomes a concern, Johnson–Lindenstrauss random projection offers a theoretically grounded reduction strategy preserving PH up to bounded distortion [5], and kNN-graph spectral distances offer an alternative to Euclidean distances more robust to ambient noise [4].

10.3 Prototype Pipeline

1. Ingest an audio file (WAV or converted-to-WAV).
2. Extract per-frame features using sliding windows (e.g., 25ms windows, 10ms hop).
3. Assemble the feature matrix: each row is one window, each column is one feature dimension.
4. Optionally apply dimensionality reduction (JL projection or PCA) if empirical testing shows benefit.
5. Compute Vietoris-Rips persistent homology (H_0 and H_1) using Ripser.
6. Convert persistence diagrams to vectorized representations (persistence images or landscapes).
7. Classify using a simple, interpretable classifier trained on labeled real/fake corpora.
8. If flagged: run dimensional ablation to identify which feature groups drive the anomaly.
9. Output: classification result + ablation report identifying the implicated property.

11. Minimal Class Project Scope

The broader research direction described above provides context, but the proposed class deliverable is deliberately narrow. A single-semester project tests the core hypothesis with the smallest defensible setup:

11.1 Dataset

ASVspoof 2019 Logical Access (LA) partition [17]. This is the most widely used benchmark for synthetic speech detection, contains multiple synthesis/voice-conversion methods, and is freely available for research.

11.2 Embedding

39-dimensional MFCCs only: 13 static coefficients + 13 Δ + 13 Δ^2. This is the minimal embedding that captures both spectral content and continuity dynamics. Additional features (formants, jitter/shimmer) are desirable extensions but not required for an initial proof of concept.

11.3 Topological Analysis

Vietoris-Rips persistent homology computing H_0 and H_1 only, using Ripser [14]. Vectorize via persistence images. This is the most standard and well-understood TDA pipeline.

11.4 Classifier

SVM or logistic regression on persistence image features. Deliberately simple to isolate the contribution of the topological features from classifier complexity.

11.5 Ablation Experiment

One targeted ablation: compare classification performance and topological signatures using all 39 dimensions versus using only the 13 static MFCCs (removing Δ and Δ²). If removing the continuity-related dimensions significantly degrades discrimination or changes the topological signatures, that provides initial evidence for the core hypothesis.

11.6 Success Criteria

Success does not require beating state-of-the-art neural detectors. The project demonstrates value if it achieves any of the following:

- Competitive or reasonable classification accuracy using only topological features derived from a physically motivated embedding.
- A clear empirical demonstration that ablating continuity-related dimensions changes the topological signature and/or classification performance in a consistent, interpretable direction.
- Evidence that topological features capture information complementary to standard classifier features—even if TDA alone is not sufficient, it may improve an ensemble.

12. Evaluation Plan

12.1 Classification Performance

Standard metrics (accuracy, precision, recall, F1, AUC) on held-out test sets from ASVspoof 2019 LA, with particular attention to false positive rate. Benchmark against at least one published baseline (e.g., LFCC + GMM, or MFCC + classifier without TDA) to understand what topological features add.

12.2 Ablation Consistency

Across the test corpus, do the same feature groups consistently drive discrimination? If the method attributes detections to inconsistent feature groups across samples, the explanatory value is undermined.

12.3 Robustness (Stretch Goal)

If time permits: performance under common audio degradations (MP3 compression, telephony bandwidth, ambient noise). Robustness benchmarks like the ADD-C dataset [18] have shown that many existing detectors degrade sharply under codec compression and channel effects. Testing whether topological features are more or less robust under these conditions would be valuable.

12.4 Cross-Generator Generalization (Stretch Goal)

Training on deepfakes from one synthesis method and testing on another. If topological features capture physical constraints rather than generator-specific artifacts, cross-generator performance should be reasonable.

13. Open Questions for Discussion

Several design decisions would benefit from expert input:

1. What is the optimal filtration strategy? Vietoris-Rips is the default, but cubical complexes on spectrogram arrays (as in [11]) or alpha complexes might offer advantages. Are there theoretical reasons to prefer one for trajectory-type data versus grid-structured data?
2. How should we handle the scale of different feature dimensions? MFCCs, F0 (Hz), jitter (%), and formants (Hz) operate on very different scales. Standardization affects distances and therefore the filtration. What normalization strategy best preserves the topological information we care about?
3. Is there a principled way to determine when adding a new dimension helps versus hurts? Beyond empirical ablation, are there theoretical tools (e.g., from persistent homology stability theorems) that could guide this a priori?
4. What statistical framework should we use for comparing persistence diagrams? Bottleneck distance, Wasserstein distance, and kernel-based methods have different properties. Which is most appropriate for comparing a single sample against a reference distribution?
5. How sensitive is the approach to sliding window parameters (window size, hop size)? These determine the density and spacing of the point cloud and could significantly affect the topological features detected.
6. Should we consider kNN-graph spectral distances instead of Euclidean distance for the Vietoris-Rips construction, given evidence [4] that these are more robust to ambient noise in moderate dimensions?

14. Summary

This proposal outlines a deepfake detection methodology that uses persistent homology not as a black-box feature extractor but as the core of an explainable framework. The key elements are:

- **Physically motivated embedding design** where each dimension corresponds to a testable property of authentic speech, grounded in speech production physics and quantified where possible by known biomechanical bounds [2, 3, 9].
- **Topological analysis via persistent homology** to detect structural anomalies in the resulting point clouds, with precedent from published TDA-on-audio work [10, 11].
- **Dimensional ablation for explanation,** allowing each detection to be traced to a specific implicated property rather than an opaque confidence score—while acknowledging the limits of what ablation can prove.
- **Modular extensibility,** enabling the framework to adapt to new deepfake methods by adding new physically motivated dimensions without rebuilding the system.

The primary research risk is empirical: whether the topological signatures of real and synthetic speech differ sufficiently in our chosen embedding to yield reliable discrimination. A minimal class project scoped to ASVspoof 2019, 39-dimensional MFCCs, Vietoris-Rips H_0/H_1, and one ablation experiment can test this core hypothesis within a semester.

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